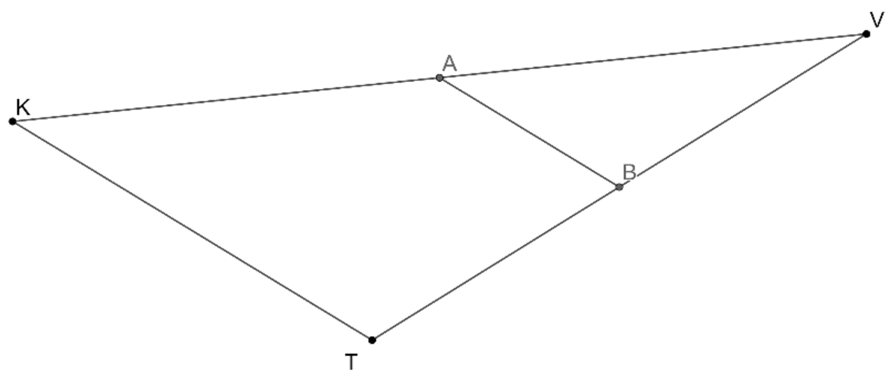
Geometry
Unit 11
Lesson 8 Practice

Name____________________
Date_________________
Period__________

Use this information to answer questions 1 through 5.

Triangle KVT is shown, where $\overline{AB}$ is the midsegment. $KT = 2x + 4$, $AB = \frac{1}{3}x + 4$, $AV = \frac{3}{2}y + 3.5$, $KA = \frac{1}{2}y + 7.5$, $VT = 5z + 3$, and $TB = \frac{3}{4}x + 6$.



1. Find the value of x.

2. Find the value of y.

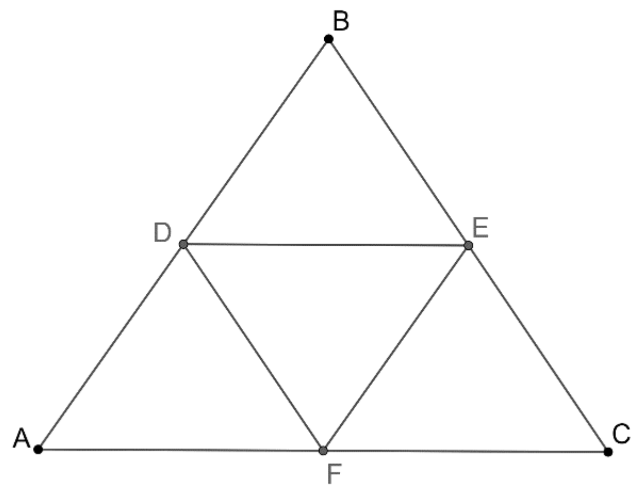
3. Find the value of z.

4. Find the length of the midsegment, $\overline{AB}$.

5. Find the length of $\overline{KV}$.

Use this information to answer questions 6 through 10.

Triangle ABC is shown, where $D, E,$ and F are midpoints. $DE = 2x - 4$, $AC = \frac{1}{2}x + 13$, $DF = \frac{3}{2}x$, $BC = 5x - 12$, $FE = 3x - 11$, and $AB = \frac{5}{3}x + 4$.



6. Find the value of x.

7. Find the length of $\overline{AB}$.

8. Find the length of $\overline{BC}$.

9. Find the length of $\overline{AC}$.

10. Find the perimeter of triangle DEF.